IILM UNIVERSITY
Greater Noida, Uttar Pradesh | School of Computer Science & Engineering

INTERNSHIP PRESENTATION

GenAI-Powered Data Analytics for Predictive Collections Strategy

Predicting customer delinquency and designing a responsible, AI-driven collections strategy for Geldium

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PROGRAM	B.Tech CSE – Specialization in AI & ML with IBM-ICE
SEMESTER	5th Semester
ORGANIZATION	Tata iQ — GenAI Powered Data Analytics Job Simulation (via Forage)

EXPERIENCE OVERVIEW

What I Worked On

Simulation

Tata iQ — GenAI Powered Data Analytics Job Simulation

Completed

August 2026

Simulated Client

Geldium — a consumer-lending company

Role

Data Analyst supporting the Collections function

THE FOUR-TASK JOURNEY

1 — TASK 1

EDA & Risk Profiling

2 — TASK 2

Predictive Modelling

3 — TASK 3

Business Reporting & Storytelling

4 — TASK 4

AI-Driven Collections Strategy

This was a virtual job simulation — not employment at Tata. All data, clients and scenarios are simulated for learning purposes.

ORGANIZATION CONTEXT

Tata, Tata iQ & Forage

TATA GROUP

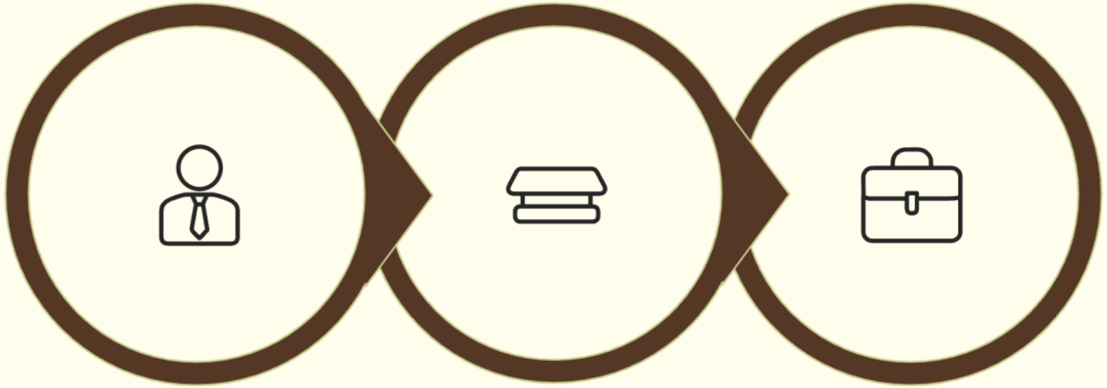
One of India's largest and most trusted diversified business conglomerates, operating across more than 100 countries in sectors spanning technology, manufacturing, financial services and consumer products.

TATA iQ

The analytics and AI arm of Tata, applying data science, machine learning and Generative AI to solve real business problems — translating raw data into strategic intelligence for enterprise clients.

FORAGE

A virtual work-experience platform that delivers employer-designed job simulations, enabling students to develop industry-relevant skills through authentic, structured business tasks.



Tata iQ Design

Forage Platform

Geldium Use Case

THE BRIEF

From Reactive Collections to Predictive Risk

Geldium's collections team lacked a systematic, data-driven way to identify at-risk customers early and prioritise outreach before accounts became seriously delinquent.

01 · COSTLY & REACTIVE

Late-stage collections are expensive and can damage customer relationships. Acting after default is far costlier than early intervention.

02 · SIGNALS MISSED EARLY

Rules-based approaches can miss the combined behavioural and financial signals that together indicate rising delinquency risk.

03 · THE ASK

Build a predictive approach that flags delinquency risk early and translates model insights into a responsible, data-driven collections strategy.

1

Predict Risk Earlier

2

Prioritise Better

3

Intervene Responsibly

PROJECT OBJECTIVES

Six Objectives

1

Assess Data Quality

Assess data quality and resolve missing values before any modelling begins.

1

Translate into Recommendations

Translate model insights into SMART, business-facing recommendations.

2

Identify Risk Indicators

Identify early behavioural and financial signals that predict delinquency risk.

2

Design Agentic AI Strategy

Design an agentic AI-driven collections strategy with human oversight built in.

3

Design Modelling Pipeline

Design a rigorous predictive modelling pipeline from feature selection to evaluation.

3

Responsible AI Principles

Apply fairness, explainability and regulatory-compliance principles throughout.

TASK 1 — DATA QUALITY

Understanding the Data

SIMULATION DATASET OVERVIEW

500
Customer Records

19
Fields per Record

16%
Base Delinquency Rate

0
Duplicate IDs

MISSING DATA IDENTIFIED

Income	7.8%
Loan Balance	5.8%
Credit Score	0.4%

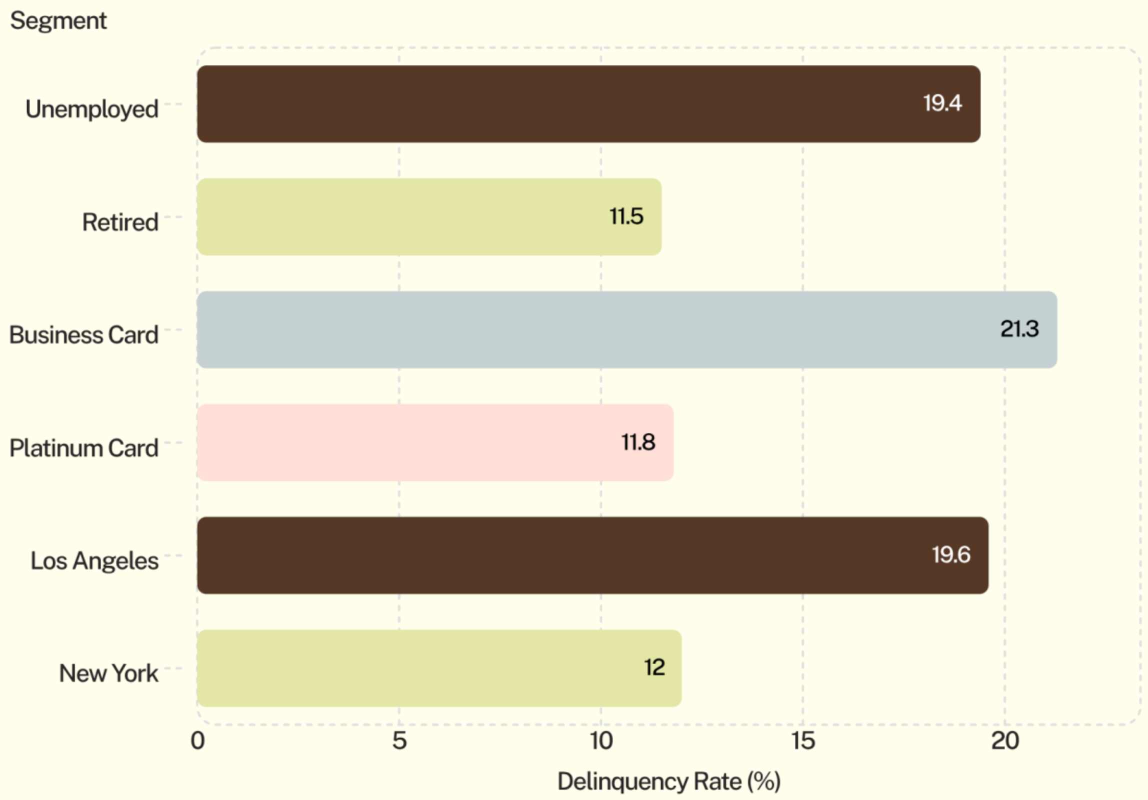
KEY QUALITY ACTIONS TAKEN

- Grouped median imputation applied to resolve missing numerical fields
- Employment Status labels standardised from **6 → 4 categories**
- **4 records** with >100% credit utilization flagged for confirmation

TASK 1 — EDA & RISK INDICATORS

What the Data Revealed

DELINQUENCY RATES BY SEGMENT



Source: Simulation dataset analysis. Segments are illustrative, not causal.

THE UNEXPECTED FINDING

⚠ WEAK UNIVARIATE CORRELATION

None of the core numerical variables showed meaningful *linear* correlation with delinquency when examined in isolation.

Implication

Risk likely depends on **non-linear interactions** across multiple variables — not simple one-variable thresholds. This justifies comparing interpretable and ensemble modelling approaches.

These findings describe patterns in the simulation dataset. They do not establish causal relationships.

TASK 2 — MODELLING APPROACH

Predictive Modelling Pipeline

MODEL SELECTION

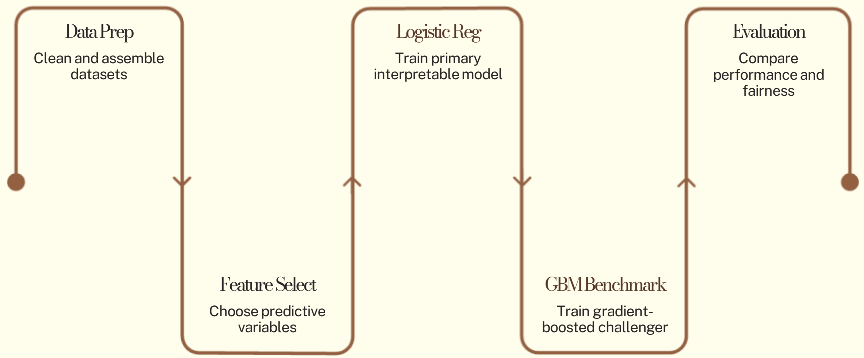
PRIMARY: Logistic Regression

- Highly interpretable coefficients
- Transparent and auditable
- Common standard in credit scoring
- Lightweight and easy to monitor

CHALLENGER: Gradient-Boosted Trees

- Captures non-linear interactions
- Suitable for complex risk patterns
- Requires additional explainability (e.g. SHAP)

MODELLING PIPELINE



TOP CANDIDATE PREDICTIVE FEATURES

01
Missed Payments

02
Credit Utilization

03
Debt-to-Income Ratio

04
6-Month Payment Trend

05
Credit Score

TASK 2 — MODEL EVALUATION

Accuracy Alone Is Not Enough

> *"With only 16% of customers delinquent, a model predicting everyone as non-delinquent could appear **84% accurate** while catching **zero** at-risk customers."*

EVALUATION METRICS

1

Precision

Limits unnecessary outreach to low-risk customers.

2

Recall

Priority metric. Missing at-risk customers is the most costly error.

3

F1 Score

Balances precision and recall into a single measure.

4

AUC-ROC

Measures model separability across all decision thresholds.

FAIRNESS & BIAS CHECKS

- Performance breakdown by **Employment Status**
- Performance breakdown by **Location**
- Performance breakdown by **Age Bands**
- Review for **proxy-variable** discrimination
- Disparate-impact ratio / **80% rule** applied
- **Threshold adjustment** if disparities appear across groups

TASK 3 — BUSINESS RECOMMENDATION

The SMART Recommendation

> **"Flag customers crossing 70% credit utilization before a payment is missed."**

S · SPECIFIC

Target customers whose credit utilization exceeds 70% for two consecutive billing cycles — before any payment is missed.

M · MEASURABLE

Compare flagged customers against a comparable control group, tracked quarterly.

A · ACTIONABLE

Route flagged accounts into an early-engagement queue for proactive, supportive outreach.

R · RELEVANT

Directly addresses a key risk signal and shifts the collections function from reactive to preventive.

T · TIME-BOUND

Run a one-quarter pilot. *Proposed* expansion criterion: delinquency drops ≥10% versus the control group.

The ≥10% reduction is the proposed expansion criterion for the pilot — not an achieved result.